DATABASE DESIGN & IMPLEMENTATION DOCUMENT

NAILVERSE

Not just NAILS.. IT's the WHOLE MOOD

Online Nail Art & Nail Beauty Online Platform

WEEK - 6

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6. Service Review and Rating Management System

Scenario

After a nail service is finished, Nailverse invites the customer to say how it went. A short comment and a star rating from 1 to 5 tell the salon what customers enjoy and which services need attention. The database therefore has to keep every opinion, calculate how well each service is doing, and point out the services that customers rate highly.

In this module a REVIEW table is added and connected to the existing CUSTOMER table (Week 1) and SERVICE table (Week 2). A review is always written by one customer about one service.

1. Create the Review Table

Review Entity

Holds one row for every opinion a customer gives about a service.

Attribute	Description
Review_ID	Unique number of the review
Customer_ID	Customer who wrote the review
Service_ID	Service being reviewed
Rating	Star rating given by the customer (1 to 5)
Review_Comment	Short feedback written by the customer
Review_Date	Day on which the review was posted

```
SQL> CREATE TABLE Review(Review_ID VARCHAR2(7),
  2  Customer_ID VARCHAR2(20),
  3  service_ID VARCHAR2(7),
  4  Rating NUMBER(1),
  5  Review_Date DATE,
  6  PRIMARY KEY (Review_ID),
  7  FOREIGN KEY(Customer_ID) REFERENCES Customer(Customer_ID),
  8  FOREIGN KEY(Service_ID) REFERENCES service(Service_ID));

Table created.

SQL> desc Review
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------
-----------
 REVIEW_ID                                 NOT NULL VARCHAR2(7)
 CUSTOMER_ID                                        VARCHAR2(20)
 SERVICE_ID                                         VARCHAR2(7)
 RATING                                             NUMBER(1)
 REVIEW_DATE                                        DATE
```

Relationships

CUSTOMER (1) ——————— (M) REVIEW – one customer can write many reviews.

SERVICE (1) ——————— (M) REVIEW – one service can collect many reviews.

2. Store Customer Feedback and Ratings

Tasks:

1. Add reviews for the services customers have received.

```
SQL> INSERT INTO Review
  2  (Review_ID, Customer_ID, Service_ID, Rating, Review_Date, R
eview_Comment)
  3  VALUES ('REV0003', 'CU-1003', 'SRV001', 4, DATE '2026-09-03
', 'Excellent service');

1 row created.

SQL>
SQL> INSERT INTO Review
  2  (Review_ID, Customer_ID, Service_ID, Rating, Review_Date, R
eview_Comment)
  3  VALUES ('REV0004', 'CU-1003', 'SRV002', 5, DATE '2026-09-04
', 'Beautiful nail art');

1 row created.

SQL>
SQL> INSERT INTO Review
  2  (Review_ID, Customer_ID, Service_ID, Rating, Review_Date, R
eview_Comment)
  3  VALUES ('REV0005', 'CU-1004', 'SRV003', 3, DATE '2026-09-05
', 'Good experience');

1 row created.
```

The other eleven reviews are added in the same way.

Review table after insertion:

```
SQL> select * from Review;

REVIEW_ CUSTOMER_ID          SERVICE     RATING REVIEW_DA
------- -------------------- ------- ---------- ---------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REV0001 CU-1001              SRV003           5 02-SEP-26
Very relaxing spa manicure

REV0003 CU-1003              SRV001           4 03-SEP-26
Excellent service

REV0004 CU-1003              SRV002           5 04-SEP-26
Beautiful nail art


REVIEW_ CUSTOMER_ID          SERVICE     RATING REVIEW_DA
------- -------------------- ------- ---------- ---------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REV0005 CU-1004              SRV003           3 05-SEP-26
Good experience

REV0006 CU-1005              SRV004           5 06-SEP-26
Loved the service

REV0007 CU-1006              SRV001           4 07-SEP-26
Friendly staff


REVIEW_ CUSTOMER_ID          SERVICE     RATING REVIEW_DA
------- -------------------- ------- ---------- ---------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REV0008 CU-1007              SRV002           5 08-SEP-26
Amazing nail design

REV0009 CU-1008              SRV003           4 09-SEP-26
Very comfortable

REV0010 CU-1009              SRV004           5 10-SEP-26
Highly recommended


REVIEW_ CUSTOMER_ID          SERVICE     RATING REVIEW_DA
------- -------------------- ------- ---------- ---------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REV0002 CU-1001              SRV003           5 02-SEP-26
Very relaxing spa manicure


10 rows selected.
```

2. Store customer ratings from 1 to 5 stars. This is ensured by adding a CHECK constraint.

```
SQL> ALTER TABLE Review ADD CONSTRAINT CHK_Rating CHECK(Rating >= 1 AND Rating <= 5);

Table altered.
```

To test the rule, a review with a rating of 6 is inserted. The database refuses it.

```
SQL> INSERT INTO Review VALUES ('REV0010', 'CU-1002', 'SRV003', 6, 'Excellent service', SYSDATE);
INSERT INTO Review VALUES ('REV0010', 'CU-1002', 'SRV003', 6, 'Excellent service', SYSDATE)
                                                           *
ERROR at line 1:
ORA-01858: a non-numeric character was found where a numeric was expected
```

3. Update a review comment if required.

```
SQL> ALTER TABLE Review ADD CONSTRAINT CHK_Rating CHECK(Rating >= 1 AND Rating <=
5);

Table altered.

SQL> UPDATE Review SET Review_Comment = 'Clean manicure, could be more detailed' W
HERE Review_ID = 'REV0008';

1 row updated.
```

4.

3. Retrieve Review Details

Write SQL queries to:

1. Display all reviews of one service (Spa Pedicure).

```
SQL> SELECT * FROM Review WHERE Service_ID = 'SRV003' ORDER BY Review_ID;

REVIEW_ CUSTOMER_ID          SERVICE     RATING REVIEW_DA
------- -------------------- ------- ---------- ---------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REV0001 CU-1001              SRV003           5 02-SEP-26
Very relaxing spa manicure

REV0002 CU-1001              SRV003           5 02-SEP-26
Very relaxing spa manicure

REV0005 CU-1004              SRV003           3 05-SEP-26
Good experience


REVIEW_ CUSTOMER_ID          SERVICE     RATING REVIEW_DA
------- -------------------- ------- ---------- ---------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REV0009 CU-1008              SRV003           4 09-SEP-26
```

2. Display each customer name with their reviews.

```
SQL> SELECT c.First_Name || ' ' || c.Last_Name AS Customer_Name, s.Service_Name, r.Rating, r.Review_Comment FROM Customer c JO
Review r ON c.Customer_ID = r.Customer_ID JOIN Service s ON r.Service_ID = s.Service_ID ORDER BY r.Review_ID;

CUSTOMER_NAME
----------------------------------------
SERVICE_NAME                                       RATING
-------------------------------------------------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
MEERA IYER
NAIL EXTENSIONS                                         5
Very relaxing spa manicure

MEERA IYER
NAIL EXTENSIONS                                         5
Very relaxing spa manicure

CUSTOMER_NAME
----------------------------------------
SERVICE_NAME                                       RATING
-------------------------------------------------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------

ANANYA  SURESH
CLASSIC MAINCURE                                        4
Excellent service

ANANYA  SURESH
CLASSIC PEDICURE                                        5

CUSTOMER_NAME
----------------------------------------
SERVICE_NAME                                       RATING
-------------------------------------------------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
Beautiful nail art

PRIYA NAIR
NAIL EXTENSIONS                                         3
Good experience

NISHA KUMARI

CUSTOMER_NAME
----------------------------------------
SERVICE_NAME                                       RATING
-------------------------------------------------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
CHROME NAIL ART                                         5
Loved the service

VARSHA RAJAN
CLASSIC MAINCURE                                        4
Friendly staff
```

```
CUSTOMER_NAME
------------------------------------
SERVICE_NAME                                RATING
------------------------------------------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------

SRI VALLI
CHROME NAIL ART                                      5
Highly recommended


10 rows selected.
```

3. Find the services that have the most reviews.

```
SQL> > SELECT s.Service_ID, s.Service_Name, COUNT(r.Review_ID) AS Review_Count FROM Service s JOIN R
eview r ON s.Service_ID = r.Service_ID GROUP BY s.Service_ID, s.Service_Name ORDER BY Review_Count D
ESC, s.Service_Name;
SP2-0734: unknown command beginning "> SELECT s..." - rest of line ignored.
SQL>
SQL> SELECT s.Service_ID,
  2         s.Service_Name,
  3         COUNT(r.Review_ID) AS Review_Count
  4  FROM Service s
  5  JOIN Review r
  6      ON s.Service_ID = r.Service_ID
  7  GROUP BY s.Service_ID, s.Service_Name
  8  ORDER BY Review_Count DESC, s.Service_Name;

SERVICE SERVICE_NAME                                   REVIEW_COUNT
------- ---------------------------------------------- ------------
SRV003  NAIL EXTENSIONS                                           4
SRV004  CHROME NAIL ART                                           2
SRV001  CLASSIC MAINCURE                                          2
SRV002  CLASSIC PEDICURE                                          2
```

4. Display the most recent customer feedback.

```
SQL> SELECT *
  2  FROM (
  3      SELECT c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  4             s.Service_Name,
  5             r.Rating,
  6             r.Review_Comment AS Customer_Feedback,
  7             r.Review_Date
  8      FROM Review r
  9      JOIN Customer c
 10          ON r.Customer_ID = c.Customer_ID
 11      JOIN Service s
 12          ON r.Service_ID = s.Service_ID
 13      ORDER BY r.Review_Date DESC
 14  )
 15  WHERE ROWNUM <= 5;

CUSTOMER_NAME
----------------------------------------
SERVICE_NAME                                       RATING
-------------------------------------------------- ----------
CUSTOMER_FEEDBACK
--------------------------------------------------------------------------------
REVIEW_DA
---------
SRI VALLI
CHROME NAIL ART                                         5
Highly recommended
10-SEP-26


CUSTOMER_NAME
----------------------------------------
SERVICE_NAME                                       RATING
-------------------------------------------------- ----------
CUSTOMER_FEEDBACK
--------------------------------------------------------------------------------
REVIEW_DA
---------
HARI KUMAR
NAIL EXTENSIONS                                         4
Very comfortable
09-SEP-26
```

```
CUSTOMER_NAME
----------------------------------------
SERVICE_NAME                                       RATING
-------------------------------------------------- ----------
CUSTOMER_FEEDBACK
--------------------------------------------------------------------------------
REVIEW_DA
---------
MOSHAMI JUN
CLASSIC PEDICURE                                        5
Clean manicure, could be more detailed
08-SEP-26


CUSTOMER_NAME
----------------------------------------
SERVICE_NAME                                       RATING
-------------------------------------------------- ----------
CUSTOMER_FEEDBACK
--------------------------------------------------------------------------------
REVIEW_DA
---------
VARSHA RAJAN
CLASSIC MAINCURE                                        4
Friendly staff
07-SEP-26


CUSTOMER_NAME
----------------------------------------
SERVICE_NAME                                       RATING
-------------------------------------------------- ----------
CUSTOMER_FEEDBACK
--------------------------------------------------------------------------------
REVIEW_DA
---------
NISHA KUMARI
CHROME NAIL ART                                         5
Loved the service
06-SEP-26
```

5. Retrieve the reviews with a rating above 4.

```
SQL> SELECT * FROM Review WHERE Rating > 4
  2  ORDER BY Review_ID;

REVIEW_ CUSTOMER_ID          SERVICE     RATING REVIEW_DA
------- -------------------- ------- ---------- ---------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REV0001 CU-1001              SRV003           5 02-SEP-26
Very relaxing spa manicure

REV0002 CU-1001              SRV003           5 02-SEP-26
Very relaxing spa manicure

REV0004 CU-1003              SRV002           5 04-SEP-26
Beautiful nail art


REVIEW_ CUSTOMER_ID          SERVICE     RATING REVIEW_DA
------- -------------------- ------- ---------- ---------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REV0006 CU-1005              SRV004           5 06-SEP-26
Loved the service

REV0008 CU-1007              SRV002           5 08-SEP-26
Clean manicure, could be more detailed

REV0010 CU-1009              SRV004           5 10-SEP-26
Highly recommended


6 rows selected.
```

4. Calculate Average Service Ratings

Aggregate functions are used to study the ratings.

Tasks:

1. Calculate the average rating of each service.

```
SQL> SELECT s.Service_Name,
  2         ROUND(AVG(r.Rating), 2) AS Average_Rating
  3  FROM Service s
  4  JOIN Review r
  5      ON s.Service_ID = r.Service_ID
  6  GROUP BY s.Service_Name
  7  ORDER BY s.Service_Name;

SERVICE_NAME                                       AVERAGE_RATING
-------------------------------------------------- --------------
CHROME NAIL ART                                                 5
CLASSIC MAINCURE                                                4
CLASSIC PEDICURE                                                5
NAIL EXTENSIONS                                              4.25
```

2. Count the number of reviews for each service.

```
SQL> SELECT s.Service_Name,
  2         COUNT(r.Rating) AS Count_Rating
  3  FROM Service s
  4  JOIN Review r
  5      ON s.Service_ID = r.Service_ID
  6  GROUP BY s.Service_Name
  7  ORDER BY s.Service_Name;

SERVICE_NAME                                       COUNT_RATING
-------------------------------------------------- ------------
CHROME NAIL ART                                               2
CLASSIC MAINCURE                                              2
CLASSIC PEDICURE                                              2
NAIL EXTENSIONS                                               4
```

3. Find the highest-rated service.

```
SQL> SELECT s.Service_ID,
  2         s.Service_Name,
  3         ROUND(AVG(r.Rating), 2) AS Average_Rating
  4  FROM Service s
  5  JOIN Review r
  6      ON s.Service_ID = r.Service_ID
  7  GROUP BY s.Service_ID, s.Service_Name
  8  HAVING AVG(r.Rating) = (
  9      SELECT MAX(Avg_Rating)
 10      FROM (
 11          SELECT AVG(Rating) AS Avg_Rating
 12          FROM Review
 13          GROUP BY Service_ID
 14      )
 15  );

SERVICE SERVICE_NAME                                       AVERAGE_RATING
------- -------------------------------------------------- --------------
SRV004  CHROME NAIL ART                                                 5
SRV002  CLASSIC PEDICURE                                                5
```

4. Identify the services whose average rating is above 4.

```
SQL> SELECT s.Service_ID,
  2         s.Service_Name,
  3         ROUND(AVG(r.Rating), 2) AS Average_Rating
  4  FROM Service s
  5  JOIN Review r
  6      ON s.Service_ID = r.Service_ID
  7  GROUP BY s.Service_ID, s.Service_Name
  8  HAVING AVG(r.Rating) > 4
  9  ORDER BY Average_Rating DESC, s.Service_Name;

SERVICE SERVICE_NAME                                       AVERAGE_RATING
------- -------------------------------------------------- --------------
SRV004  CHROME NAIL ART                                                 5
SRV002  CLASSIC PEDICURE                                                5
SRV003  NAIL EXTENSIONS                                              4.25
```

5. Identify Highly Rated Services

Prepare service rating reports.

Report 1: Service Rating Analysis

Display:

- Service name
- Number of reviews
- Average rating

A LEFT JOIN is used so that services which have not been reviewed yet are also listed.

```
SQL> SELECT s.Service_Name,
  2         COUNT(r.Review_ID) AS Number_of_Reviews,
  3         ROUND(AVG(r.Rating), 2) AS Average_Rating
  4  FROM Service s
  5  LEFT JOIN Review r
  6      ON s.Service_ID = r.Service_ID
  7  GROUP BY s.Service_ID, s.Service_Name
  8  ORDER BY Average_Rating DESC NULLS LAST, s.Service_Name;

SERVICE_NAME                                       NUMBER_OF_REVIEWS
-------------------------------------------------- -----------------
AVERAGE_RATING
--------------
CHROME NAIL ART                                                    2
             5

CLASSIC PEDICURE                                                   2
             5

NAIL EXTENSIONS                                                    4
          4.25


SERVICE_NAME                                       NUMBER_OF_REVIEWS
-------------------------------------------------- -----------------
AVERAGE_RATING
--------------
CLASSIC MAINCURE                                                   2
             4

Chrome Nail Art                                                    0


SPA PEDICURE                                                       0



6 rows selected
```

Report 2: Customer Feedback Analysis

Find:

- Most reviewed services

```
SQL> SELECT *
  2  FROM (
  3      SELECT s.Service_Name,
  4             COUNT(r.Review_ID) AS Review_Count
  5      FROM Service s
  6      JOIN Review r
  7          ON s.Service_ID = r.Service_ID
  8      GROUP BY s.Service_ID, s.Service_Name
  9      ORDER BY Review_Count DESC, s.Service_Name
 10  )
 11  WHERE ROWNUM <= 3;

SERVICE_NAME                                     REVIEW_COUNT
------------------------------------------------ ------------
NAIL EXTENSIONS                                             4
CHROME NAIL ART                                             2
CLASSIC MAINCURE                                            2
```

- Highly rated services (average rating above 4)

```
SQL> SELECT s.Service_Name,
  2         ROUND(AVG(r.Rating), 2) AS Average_Rating
  3  FROM Service s
  4  JOIN Review r
  5      ON s.Service_ID = r.Service_ID
  6  GROUP BY s.Service_ID, s.Service_Name
  7  HAVING AVG(r.Rating) > 4
  8  ORDER BY Average_Rating DESC, s.Service_Name;

SERVICE_NAME                                       AVERAGE_RATING
-------------------------------------------------- --------------
CHROME NAIL ART                                                 5
CLASSIC PEDICURE                                                5
NAIL EXTENSIONS                                              4.25
```

- Services requiring improvement (average rating below 4)

```
SQL> SELECT s.Service_Name,
  2         COUNT(r.Review_ID) AS Review_Count,
  3         ROUND(AVG(r.Rating), 2) AS Average_Rating
  4  FROM Service s
  5  JOIN Review r
  6      ON s.Service_ID = r.Service_ID
  7  GROUP BY s.Service_ID, s.Service_Name
  8  ORDER BY Average_Rating, s.Service_Name;

SERVICE_NAME                                       REVIEW_COUNT AVERAGE_RATING
-------------------------------------------------- ------------ --------------
CLASSIC MAINCURE                                              2              4
NAIL EXTENSIONS                                               4           4.25
CHROME NAIL ART                                               2              5
CLASSIC PEDICURE                                              2              5
```

Report 3: Rating Distribution

- Number of 5-star ratings

```
SQL> SELECT COUNT(*) AS Five_Star_Ratings
  2  FROM Review
  3  WHERE Rating = 5;

FIVE_STAR_RATINGS
-----------------
                6
```

- Number of 4-star ratings

```
SQL> SELECT COUNT(*) AS Five_Star_Ratings
  2  FROM Review
  3  WHERE Rating = 4;

FIVE_STAR_RATINGS
-----------------
                3
```

- Number of low-rated services (average rating below 3.5)

```
SQL> SELECT COUNT(*) AS Low_Rated_Services FROM (SELECT Service_ID FROM Review GROUP BY Service_ID H
AVING AVG(Rating) < 3.5);

LOW_RATED_SERVICES
------------------
                 0
```

- Complete rating distribution

```
SQL> SELECT Rating, COUNT(*) AS Number_of_Ratings FROM Review GROUP BY Rating ORDER BY Rating DESC;

    RATING NUMBER_OF_RATINGS
---------- -----------------
         5                 6
         4                 3
         3                 1
```